✅ **Project:** `crud_project`

✅ **App:** `students`

CRUD on student records with the fields: `name`, `roll`, `email`.

Step-by-Step Setup

1. Create the Django project and app

```
django-admin startproject crud_project
cd crud_project
python manage.py startapp students
```

2. Update settings

In `crud_project/settings.py`:

```
INSTALLED_APPS = [ ...
    'students',]
```

3. Model

In `students/models.py`:

```
from django.db import models

class Student(models.Model):
    name = models.CharField(max_length=100)
    roll = models.IntegerField()
    email = models.EmailField()

    def __str__(self):
        return self.name
```

Run migrations:

```
python manage.py makemigrations
python manage.py migrate
```

4. URLs

In `crud_project/urls.py`:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [path('', include('students.urls'))]
```

In `students/urls.py` (create it):

```
from django.urls import path
from . import views

urlpatterns = [    path('', views.student_list, name='student_list'),
    path('add/', views.add_student, name='add_student'),
    path('edit/<int:id>/', views.edit_student, name='edit_student'),
    path('delete/<int:id>/', views.delete_student,
name='delete_student'),]
```

5. Views

In `students/views.py`:

```
from django.shortcuts import render, redirect, get_object_or_404
from .models import Student

def student_list(request):
    students = Student.objects.all()
    return render(request, 'students/list.html', {'students': students})

def add_student(request):
    if request.method == 'POST':
        name = request.POST['name']
        roll = request.POST['roll']
        email = request.POST['email']
        Student.objects.create(name=name, roll=roll, email=email)
        return redirect('student_list')
    return render(request, 'students/add.html')

def edit_student(request, id):
    student = get_object_or_404(Student, pk=id)
    if request.method == 'POST':
        student.name = request.POST['name']
        student.roll = request.POST['roll']
        student.email = request.POST['email']
        student.save()
        return redirect('student_list')
    return render(request, 'students/edit.html', {'student': student})

def delete_student(request, id):
    student = get_object_or_404(Student, pk=id)
    student.delete()
    return redirect('student_list')
```

6. Templates

Create folder: `students/templates/students/`

- list.html

```
<h2>Student List</h2>
<a href="{% url 'add_student' %}">Add Student</a>
<table border="1">
    <tr>
        <th>Name</th><th>Roll</th><th>Email</th><th>Actions</th>
    </tr>
    {% for student in students %}
    <tr>
        <td>{{ student.name }}</td>
        <td>{{ student.roll }}</td>
        <td>{{ student.email }}</td>
        <td>
            <a href="{% url 'edit_student' student.id %}">Edit</a> |
            <a href="{% url 'delete_student' student.id %}">Delete</a>
        </td>
    </tr>
    {% endfor %}
</table>
```

- add.html

```
<h2>Add Student</h2>
<form method="POST">
    {% csrf_token %}
    Name: <input type="text" name="name"><br><br>
    Roll: <input type="number" name="roll"><br><br>
    Email: <input type="email" name="email"><br><br>
    <input type="submit" value="Add">
</form>
<a href="{% url 'student_list' %}">Back to List</a>
```

- edit.html

```
<h2>Edit Student</h2>
<form method="POST">
    {% csrf_token %}
    Name: <input type="text" name="name" value="{{ student.name
}}"><br><br>
    Roll: <input type="number" name="roll" value="{{ student.roll
}}"><br><br>
    Email: <input type="email" name="email" value="{{ student.email
}}"><br><br>
    <input type="submit" value="Update">
</form>
<a href="{% url 'student_list' %}">Back to List</a>
```

7. Run Server

```
python manage.py runserver
```